SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1

Given Network

192.168.20.0/24

The administrator allocates:

- Lab A → /26
- Lab B → /27
- Lab C → /28

Subnet allocation is done from the beginning of the network.

Lab	Subnet Mask	Network Address	Host Range	Broadcast Address	Usable Hosts
Lab A	/26 (255.255.255.192)	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	**62**
Lab B	/27 (255.255.255.224)	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	**30**
Lab C	/28 (255.255.255.240)	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	**14**

Calculation

Lab A (/26)

- Total addresses = 64
- Usable hosts = 64 − 2 = **62**

Lab B (/27)

- Total addresses = 32
- Usable hosts = 32 − 2 = **30**

Lab C (/28)

- Total addresses = 16
- Usable hosts = 16 − 2 = **14**

Final Answer

- Lab A → 62 hosts
- Lab B → 30 hosts
- Lab C → 14 hosts

Question 2

Given

System 1

- IP = 192.168.1.130
- Mask = /26

System 2

- IP = 192.168.1.190
- Mask = /26

A /26 subnet has blocks of **64**.

Subnets are

Network	Broadcast
192.168.1.0	63
192.168.1.64	127
192.168.1.128	191
192.168.1.192	255

System 1

IP = **192.168.1.130**

Falls between **128–191**

Network Address

192.168.1.128

Broadcast

192.168.1.191

Host Range

192.168.1.129 – 192.168.1.190

Usable Hosts

62

System 2

IP = **192.168.1.190**

Also falls in

192.168.1.128–191

Network Address

192.168.1.128

Broadcast

192.168.1.191

Host Range

192.168.1.129 – 192.168.1.190

Usable Hosts

62

Final Table

System	Network	Host Range	Broadcast	Valid?	Usable Hosts
System 1	192.168.1.128	129–190	191	Yes	62
System 2	192.168.1.128	129–190	191	Yes	62

Question 3

Given Network

172.16.0.0/16

Departments

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

Allocate sequentially.

Department	Subnet	Network Address	Host Range	Broadcast	Usable Hosts
HR	/25	172.16.0.0	172.16.0.1–172.16.0.126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129–172.16.0.190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193–172.16.0.222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225–172.16.0.238	172.16.0.239	14

Remaining Unused Addresses

After Admin,

Remaining

172.16.0.240 – 172.16.255.255

Final Answer

Department	Hosts
HR	126
Sales	62
IT	30
Admin	14

Unused range

172.16.0.240 – 172.16.255.255

Question 4

Given

Network

172.16.10.0/24

Subnet Mask

/27

Step 1

Number of subnets

Borrowed bits

$27-24 = 3$

Number of subnets

$2^3 =$ **8**

Step 2

Hosts per subnet

$32-27 = 5$ host bits

$2^5 = 32$

Usable

$32-2 =$ **30**

Step 3

Subnets

Subnet	Range
0	0–31
1	32–63
2	64–95
3	96–127
4	128–159
5	160–191
6	192–223
7	224–255

IP

172.16.10.78

Falls in

64–95

Network Address

172.16.10.64

Broadcast

172.16.10.95

Host Range

172.16.10.65–172.16.10.94

Check

172.16.10.95

This is the **broadcast address** of the subnet.

Therefore,

It is **not a usable host**, although it belongs to the same subnet.

Final Answer

Item	Answer
Subnets	8
Usable Hosts/Subnet	30
Network Address	172.16.10.64
Broadcast	172.16.10.95
Host Range	65–94
172.16.10.95	Broadcast Address (Not Usable)

Question 5

IPv6 Address

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Compressed Form

Rules

- Remove leading zeros.
- Replace longest continuous zero groups with ::

Compressed

```
2001:db8::ff00:42:8329
```

Expanded Form

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Network Prefix (/64)

First 64 bits

```
2001:0db8:0000:0000::/64
```

Number of Hosts

Host bits

$128-64 = 64$

Hosts

```
2^64
=
18,446,744,073,709,551,616
```

possible interface identifiers (addresses).